Restricting and Sorting Data

1. Display the Ename, Sal, Comm from EMP table who earn commission and sort the records in descending order of Salary and Comm. Use column's numeric position in the ORDER BY clause.

```
SQL> SELECT ENAME, SAL, COMM
  2  FROM EMP
  3  WHERE COMM IS NOT NULL
  4  ORDER BY 2 DESC, 3 DESC;

ENAME                                                   SAL       COMM
-------------------------------------------------- ---------- ----------
ALLEN                                                  1600        300
TURNER                                                 1500          0
MARTIN                                                 1250       1400
WARD                                                   1250        500
```

2. The HR department needs a query to display all unique job codes from the EMP table.

```
SQL> SELECT DISTINCT JOB
  2  FROM EMP;

JOB
--------------------------------------------------
PRESIDENT
MANAGER
ANALYST
CLERK
SALESMAN
```

3. The HR department has requested a report of all employees and their job IDs. Display the **last name concatenated with the job ID** (separated by a comma and space) and name the column **Employee and Title** by giving **Column Alias**.

```
SQL> select empno as"Emp#",ename as"Employee", job as "job", hiredate as "hire date" from emp;

      Emp# Employee                                           job
---------- -------------------------------------------------- --------------------------------------------------
hire date
---------
      7839 KING                                               PRESIDENT
17-NOV-81

      7698 BLAKE                                              MANAGER
01-MAY-81

      7782 CLARK                                              MANAGER
09-JUN-81

      7566 JONES                                              MANAGER
02-APR-81

      7788 SCOTT                                              ANALYST
13-JUL-87

      7902 FORD                                               ANALYST
03-DEC-81

      7369 SMITH                                              CLERK
17-DEC-80

      7499 ALLEN                                              SALESMAN
20-FEB-81

      7521 WARD                                               SALESMAN
22-FEB-81

      7654 MARTIN                                             SALESMAN
28-SEP-81
```

4. The HR department has requested a report of all employees and their job IDs. Display the **last name concatenated with the job ID** (separated by a comma and space) and name the column **Employee and Title** by giving **Column Alias**.

```
SQL> SELECT ENAME || ',' || JOB || ',' || HIREDATE || ',' || MGR AS THE_OUTPUT
  2  from emp;

THE_OUTPUT
----------------------------------------------------------------------
KING,PRESIDENT,17-NOV-81,
BLAKE,MANAGER,01-MAY-81,7839
CLARK,MANAGER,09-JUN-81,7839
JONES,MANAGER,02-APR-81,7839
SCOTT,ANALYST,13-JUL-87,7566
FORD,ANALYST,03-DEC-81,7566
SMITH,CLERK,17-DEC-80,7902
ALLEN,SALESMAN,20-FEB-81,7698
WARD,SALESMAN,22-FEB-81,7698
MARTIN,SALESMAN,28-SEP-81,7698
TURNER,SALESMAN,08-SEP-81,7698
ADAMS,CLERK,23-MAY-87,7788
JAMES,CLERK,03-DEC-81,7698
MILLER,CLERK,23-JAN-82,7782

14 rows selected.
```

5. To familiarize yourself with the data in the **EMP** table, create a query to display all the data from that table. Separate each column output by a comma. **ENAME, JOB, HIREDATE, MGR.** Name the column title **THE_OUTPUT**.

```
SQL> SELECT ENAME, JOB, HIREDATE, MGR
  2  FROM EMP;

ENAME                                              JOB                                                HIREDATE         MGR
-------------------------------------------------- -------------------------------------------------- --------- ----------
KING                                               PRESIDENT                                          17-NOV-81
BLAKE                                              MANAGER                                            01-MAY-81       7839
CLARK                                              MANAGER                                            09-JUN-81       7839
JONES                                              MANAGER                                            02-APR-81       7839
SCOTT                                              ANALYST                                            13-JUL-87       7566
FORD                                               ANALYST                                            03-DEC-81       7566
SMITH                                              CLERK                                              17-DEC-80       7902
ALLEN                                              SALESMAN                                           20-FEB-81       7698
WARD                                               SALESMAN                                           22-FEB-81       7698
MARTIN                                             SALESMAN                                           28-SEP-81       7698
TURNER                                             SALESMAN                                           08-SEP-81       7698
ADAMS                                              CLERK                                              23-MAY-87       7788
JAMES                                              CLERK                                              03-DEC-81       7698
MILLER                                             CLERK                                              23-JAN-82       7782

14 rows selected.
```

6. Create a report to display the **ename, job, and hiredate** for the employees whose name is **SCOTT** or **TURNER**. Order the query in **ascending order by hiredate**.

```
SQL> SELECT ENAME, JOB, HIREDATE FROM EMP WHERE ENAME IN ('SCOTT','TURNER') ORDER BY HIREDATE ASC;

ENAME                                              JOB                                                HIREDATE
-------------------------------------------------- -------------------------------------------------- ---------
TURNER                                             SALESMAN                                           08-SEP-81
SCOTT                                              ANALYST                                            13-JUL-87
```

7. Display the **ename** and **department number** of all employees in departments **20 or 30** in **ascending alphabetical order by ename**.

```
SQL> SELECT ENAME, DEPTNO FROM EMP WHERE DEPTNO IN (20,30) ORDER BY ENAME ASC;

ENAME                                                  DEPTNO
-------------------------------------------------- ----------
ADAMS                                                      20
ALLEN                                                      30
BLAKE                                                      30
FORD                                                       20
JAMES                                                      30
JONES                                                      20
MARTIN                                                     30
SCOTT                                                      20
SMITH                                                      20
TURNER                                                     30
WARD                                                       30

11 rows selected.
```

8. Modify the previous query to display the **last name** and **salary** of employees who earn **between 2000 and 3000** and are in department **20 or 30**. Label the columns **Employee** and **Monthly Salary**, respectively, using **Column Alias**.

```
SQL> SELECT ENAME AS "Employee", SAL AS "Monthly Salary" FROM EMP WHERE SAL BETWEEN 2000 AND 3000 AND DEPTNO IN (20,30);

Employee                                           Monthly Salary
-------------------------------------------------- --------------
BLAKE                                                        2850
JONES                                                        2975
SCOTT                                                        3000
FORD                                                         3000
```

9. The HR department needs a report that displays the **last name** and **hire date** for all employees who were **hired in 1981**.

```
SQL> SELECT ENAME, HIREDATE FROM EMP WHERE HIREDATE BETWEEN TO_DATE('01-JAN-1981','DD-MON-YYYY') AND TO_DATE('31-DEC-1981','DD-MON-YY
YY');

ENAME                                              HIREDATE
-------------------------------------------------- ---------
KING                                               17-NOV-81
BLAKE                                              01-MAY-81
CLARK                                              09-JUN-81
JONES                                              02-APR-81
FORD                                               03-DEC-81
ALLEN                                              20-FEB-81
WARD                                               22-FEB-81
MARTIN                                             28-SEP-81
TURNER                                             08-SEP-81
JAMES                                              03-DEC-81

10 rows selected.
```

10. Display the **Ename** and **Sal** of employees who earn **more than an amount the user specifies after a Prompt**.

```
SQL> SELECT ENAME, SAL FROM EMP WHERE SAL > &SALARY;
Enter value for salary: 2000
old   1: SELECT ENAME, SAL FROM EMP WHERE SAL > &SALARY
new   1: SELECT ENAME, SAL FROM EMP WHERE SAL > 2000

ENAME                                                     SAL
-------------------------------------------------- ----------
KING                                                     5000
BLAKE                                                    2850
CLARK                                                    2450
JONES                                                    2975
SCOTT                                                    3000
FORD                                                     3000

6 rows selected.
```

11. Create a report to display the **last name** and **job title** of all employees who **do not have a manager**.

```
SQL> SELECT ENAME, JOB FROM EMP WHERE MGR IS NULL;

ENAME                                              JOB
-------------------------------------------------- --------------------------------------------------
KING                                               PRESIDENT
```

12. Create a query that prompts the user for **Manager ID** and generates **EMPNO, ENAME, SAL, DEPTNO**. The user should have the ability to **sort the records on a selected column**.

```
SQL> SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN;
Enter value for mgr: 7698
Enter value for column: ename
old   1: SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN
new   1: SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=7698 ORDER BY ename

     EMPNO ENAME                                                     SAL     DEPTNO
---------- -------------------------------------------------- ---------- ----------
      7499 ALLEN                                                    1600         30
      7900 JAMES                                                     950         30
      7654 MARTIN                                                   1250         30
      7844 TURNER                                                   1500         30
      7521 WARD                                                     1250         30
```

13. The HR department wants to run reports based on a manager. Create a query that prompts the user for a **MGR** and generates the **empno, ename, salary, and department** for that manager's employees. The HR department wants the ability to **sort the report on a selected column**.

```
SQL> SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN;
Enter value for mgr: 7698
Enter value for column: ENAME
old   1: SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN
new   1: SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=7698 ORDER BY ENAME

     EMPNO ENAME                                                     SAL     DEPTNO
---------- -------------------------------------------------- ---------- ----------
      7499 ALLEN                                                    1600         30
      7900 JAMES                                                     950         30
      7654 MARTIN                                                   1250         30
      7844 TURNER                                                   1500         30
      7521 WARD                                                     1250         30
```

14. Display all employee **last names** in which the **third letter of the name is A**.

```
SQL> SELECT ENAME FROM EMP WHERE ENAME LIKE '__A%';

ENAME
--------------------------------------------------
BLAKE
CLARK
ADAMS
```

15. Display the **last name** of all employees who have **both an A and an S** in their **ename**.

```
SQL> SELECT ENAME FROM EMP WHERE ENAME LIKE '%A%' AND ENAME LIKE '%S%';

ENAME
--------------------------------------------------
ADAMS
JAMES
```

16. Display the **Ename, Job, Sal** for all employees whose **jobs are CLERK** and whose **salary is 800 or 950 or 1300**.

```
SQL> SELECT ENAME, JOB, SAL FROM EMP WHERE JOB='CLERK' AND SAL IN (800,950,1300);

ENAME                                              JOB                                                       SAL
-------------------------------------------------- -------------------------------------------------- ----------
SMITH                                              CLERK                                                     800
JAMES                                              CLERK                                                     950
MILLER                                             CLERK                                                    1300
```